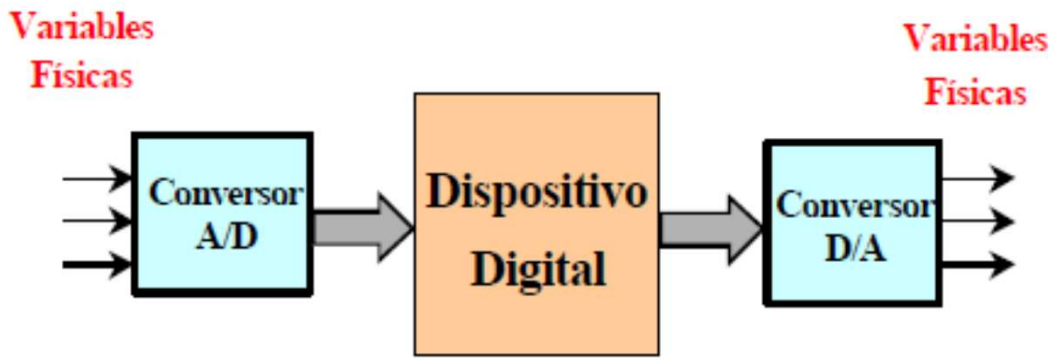
UNIVERSIDAD DE ATACAMA

AUTOMATICA Y ROBOTOCA
(Adquisición y distribución de datos).

Nahur M. Meléndez Araya.
Magister en Ingeniería. Informática.

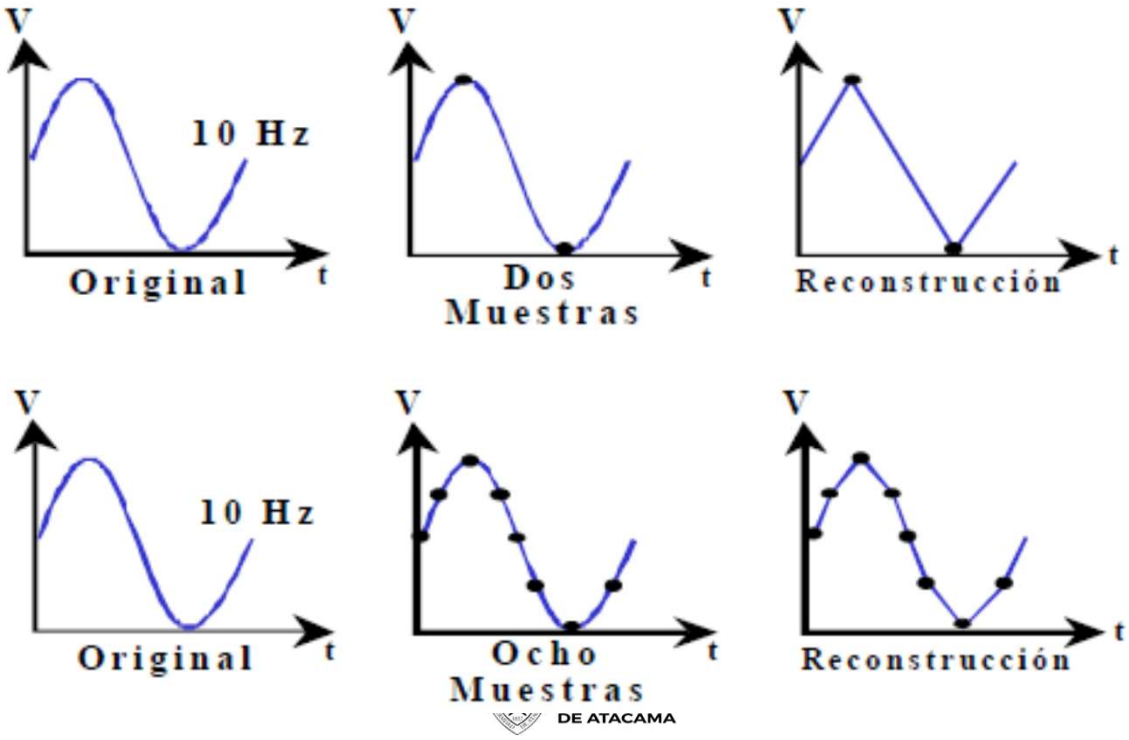
EL CONCEPTO.

- Los seres humanos procesamos nuestra información numérica utilizando el sistema decimal.

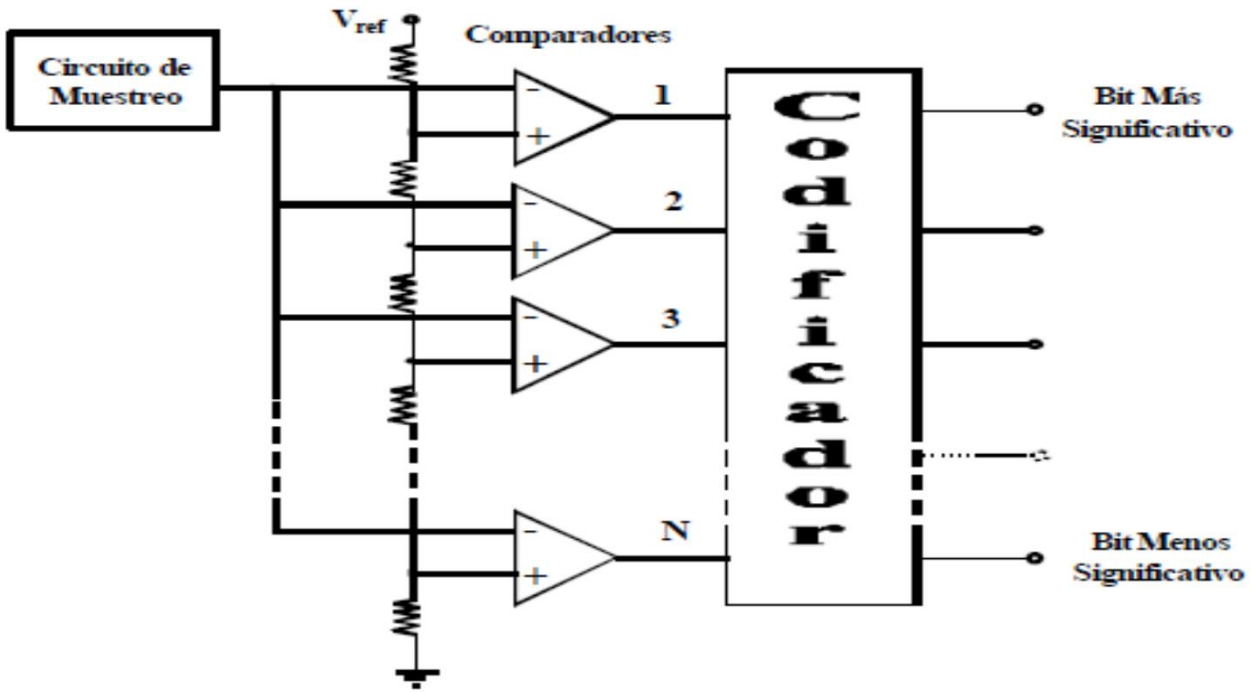


- Sin embargo, la tecnología electrónica emplea un camino diferente para manejar a muy alta velocidad y eficiencia, las mismas cantidades..

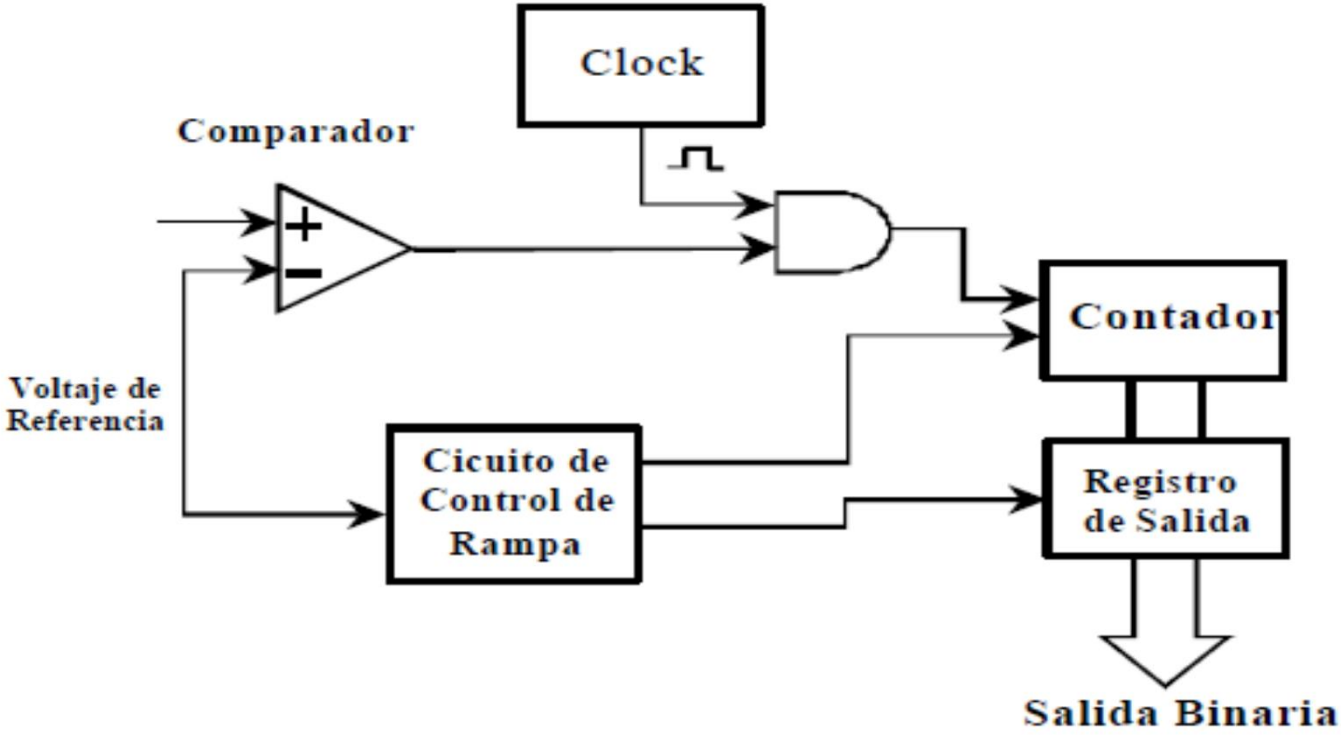
Conversión Análoga/Digital: Teoría del muestreo.



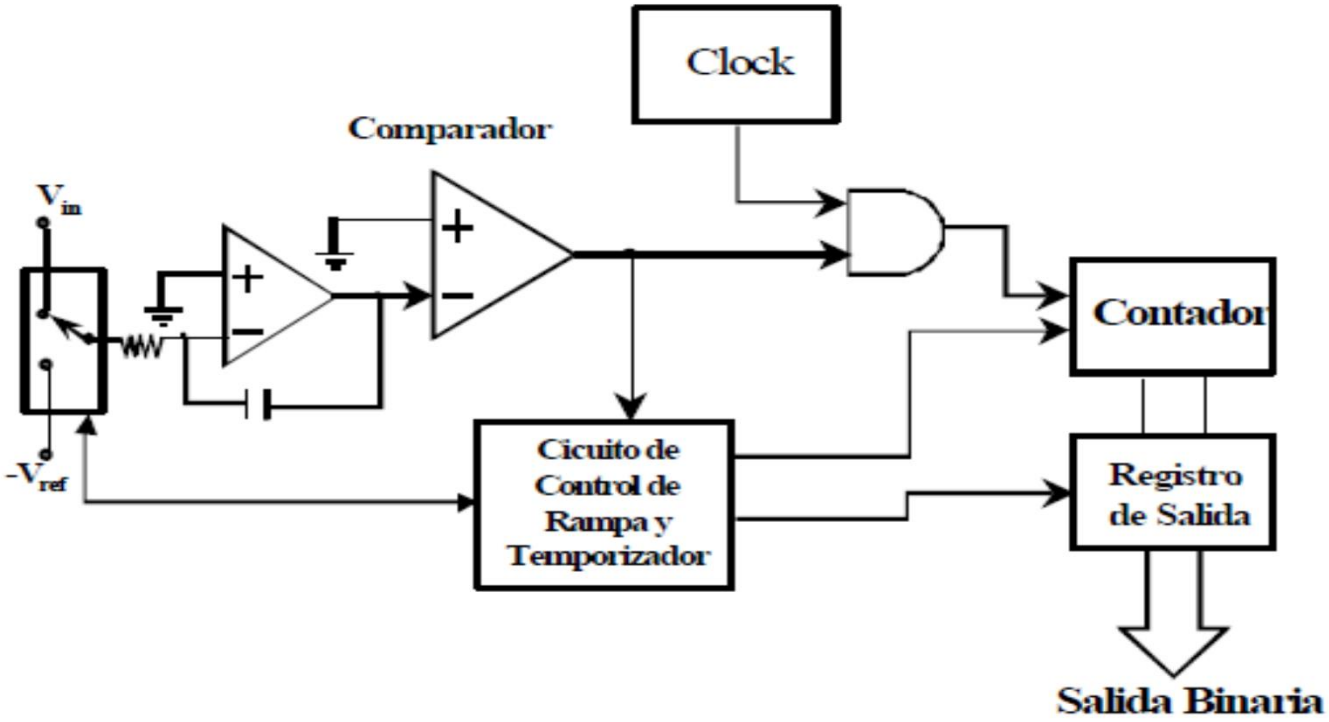
Conversión Análoga/Digital: Conversor tipo FLASH



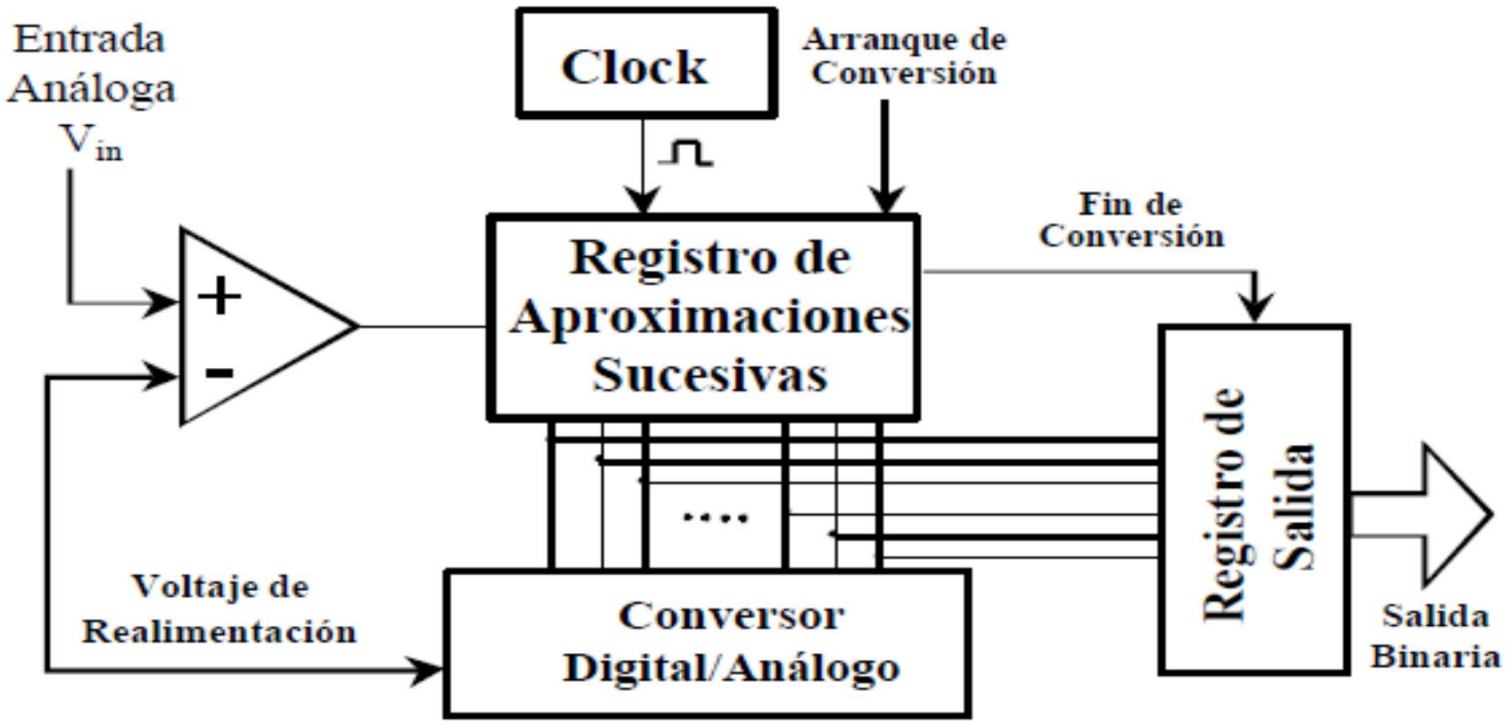
Conversión Análoga/Digital: Conversor de rampa única



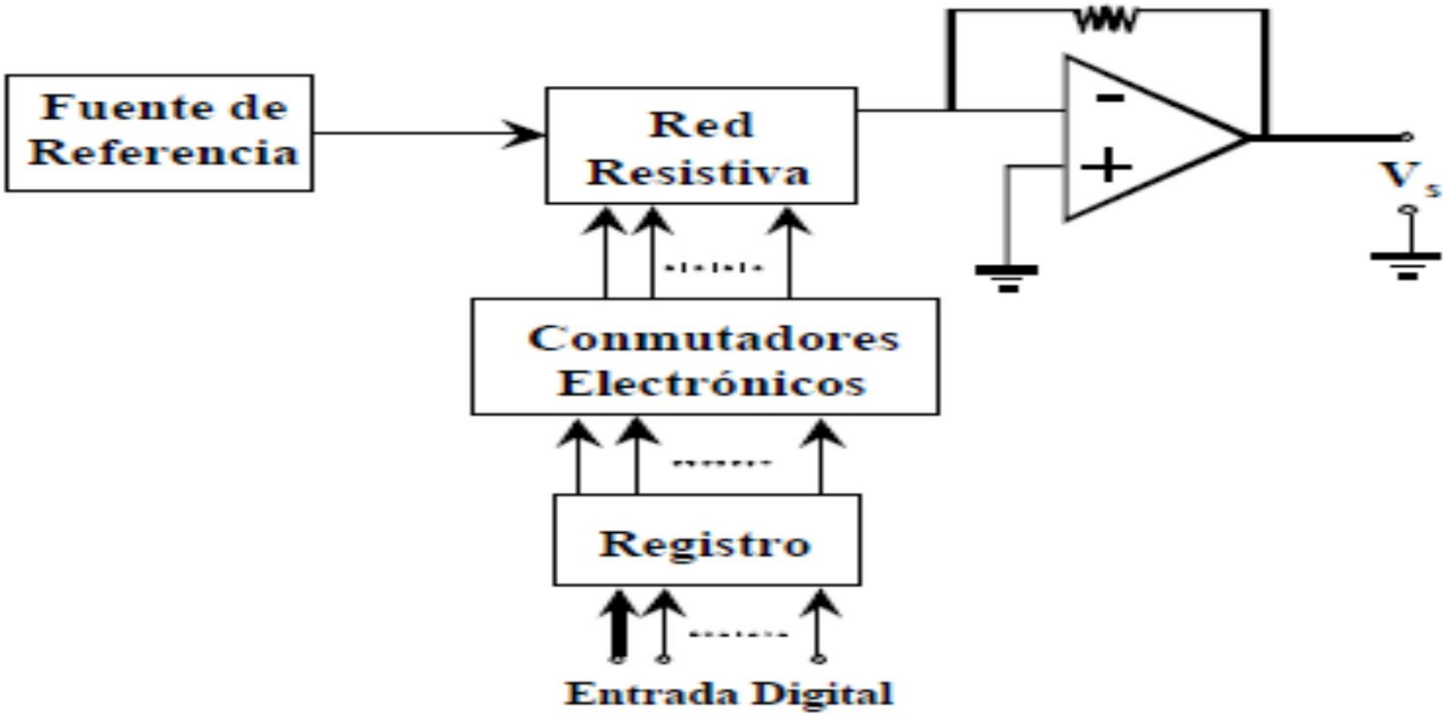
Conversión Análoga/Digital: Conversor de rampa doble



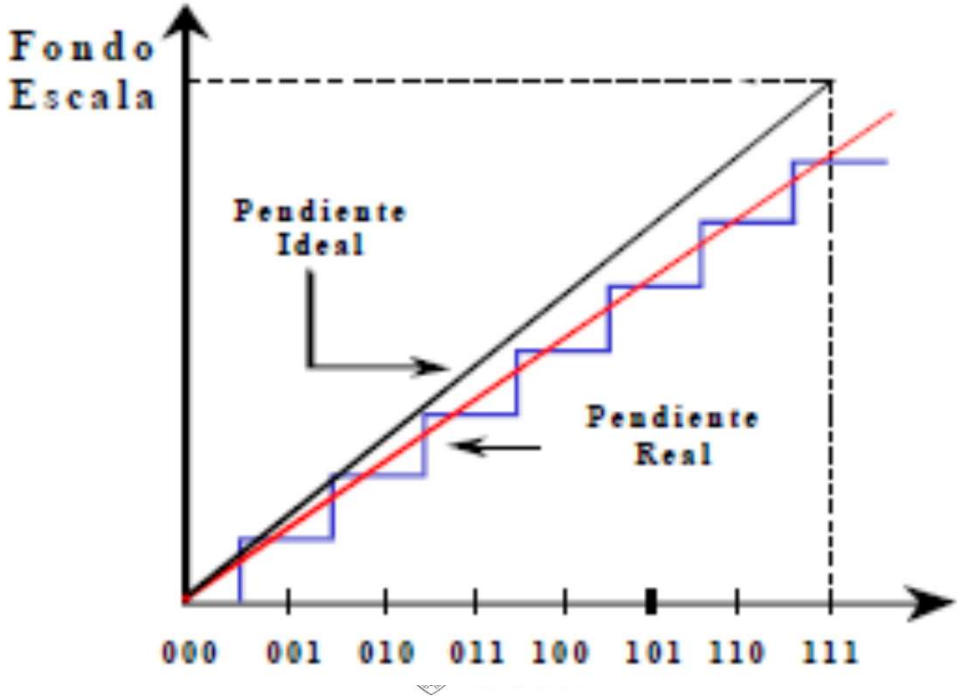
Conversión Análoga/Digital: Conversor de aproximaciones sucesivas



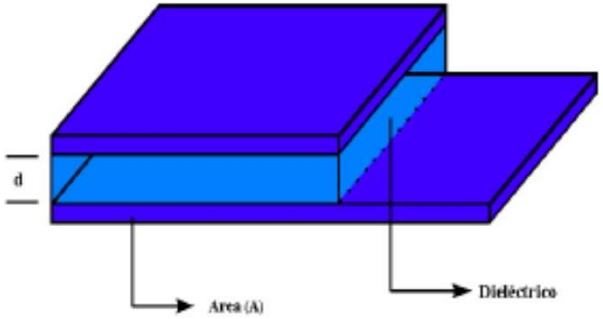
Conversión Digital/Análoga

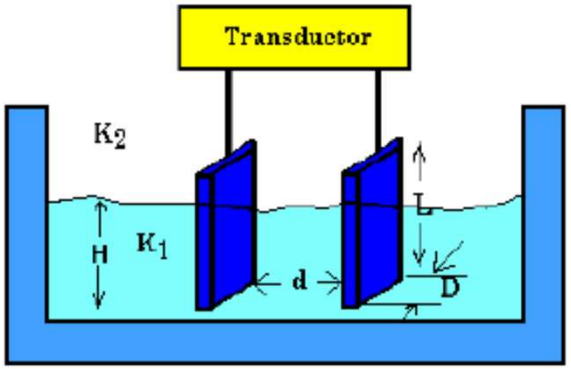


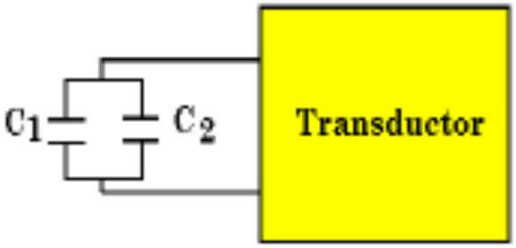
Conversión Digital/Análoga: Error de Offset y Error de Ganancia



Sensores de Nivel: Transductores de Efecto Capacitivo

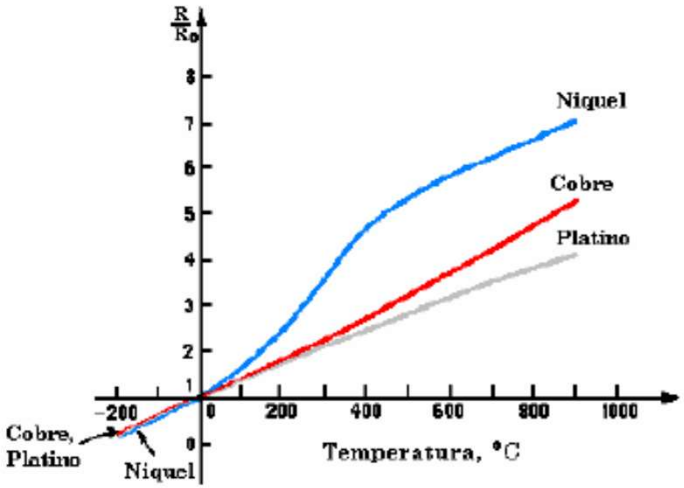




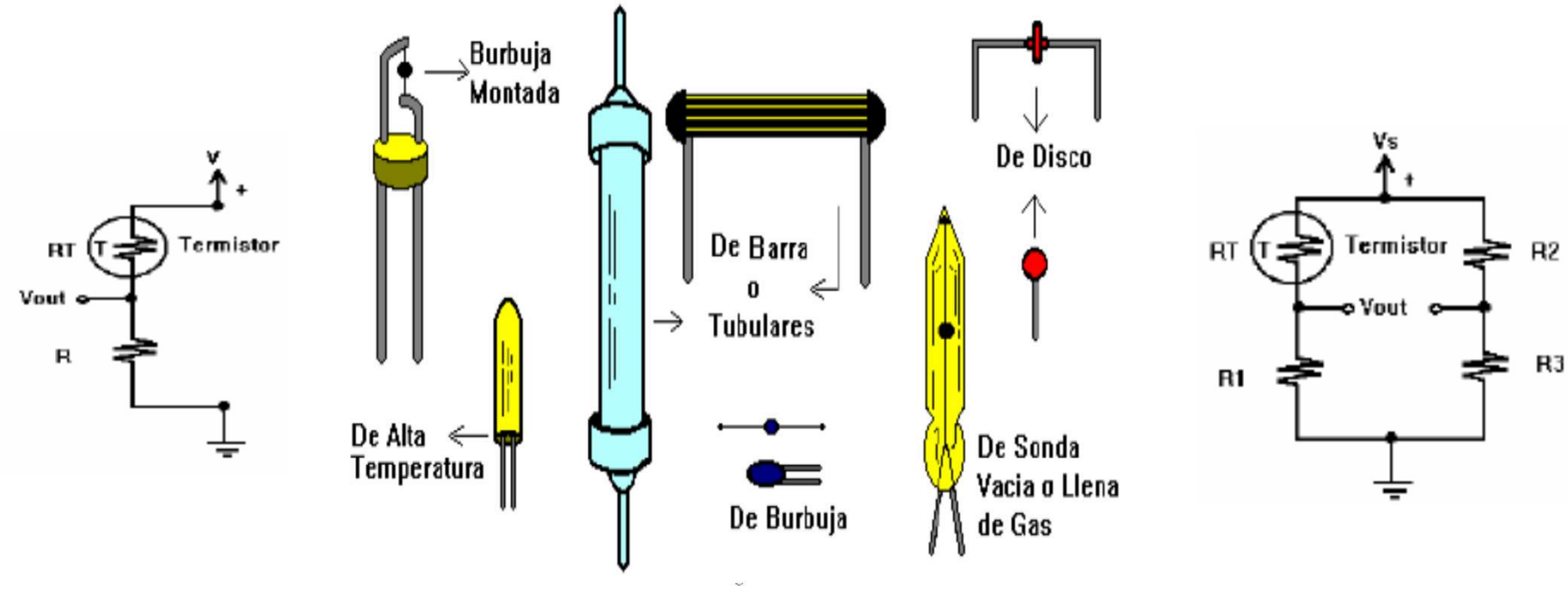


Transductores de Temperatura: Termoresistencias

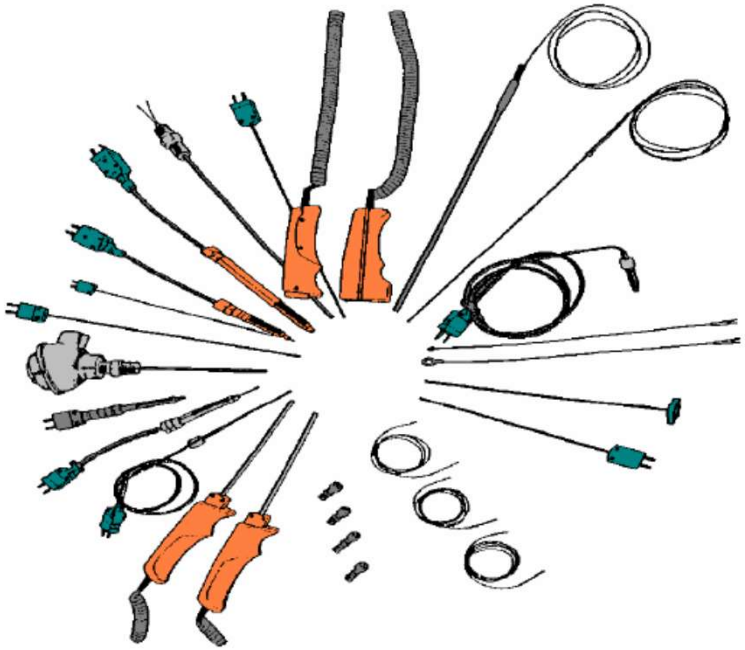
MATERIAL	COEF. DE ºT , "a" (? / ? ºC)	RESISTENCIA ESPECÍFICA (? / cm^3)	R_0 (?)	TEMPERATURA ºC
Platino	0,00385	9,83	25,5	1200
Níquel	0,0066	6,38	1..1000	-70..150
Cobre	0,00425	1,56	25	-200..150

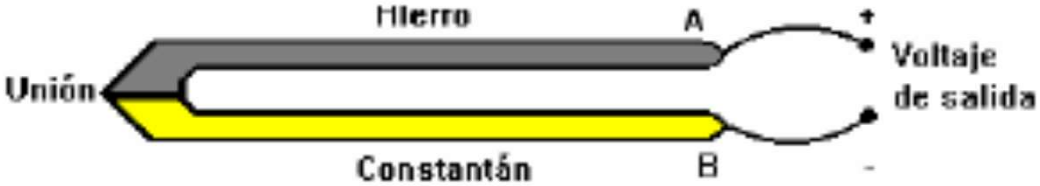


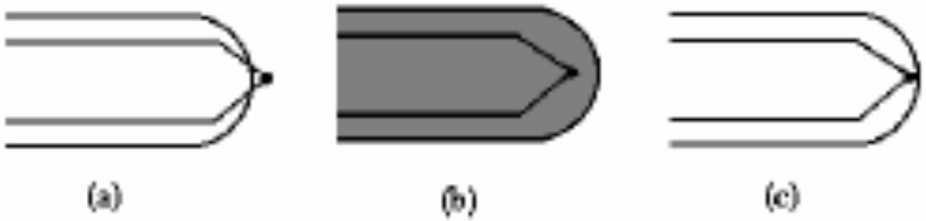
Transductores de Temperatura: Termistores NTC



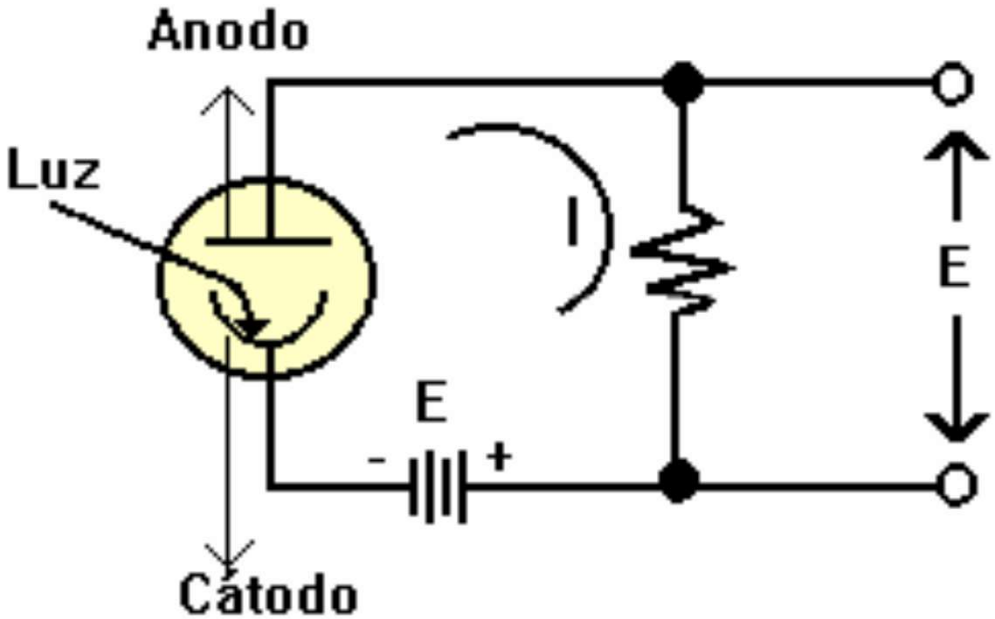
Transductores de Temperatura: Termocuplas



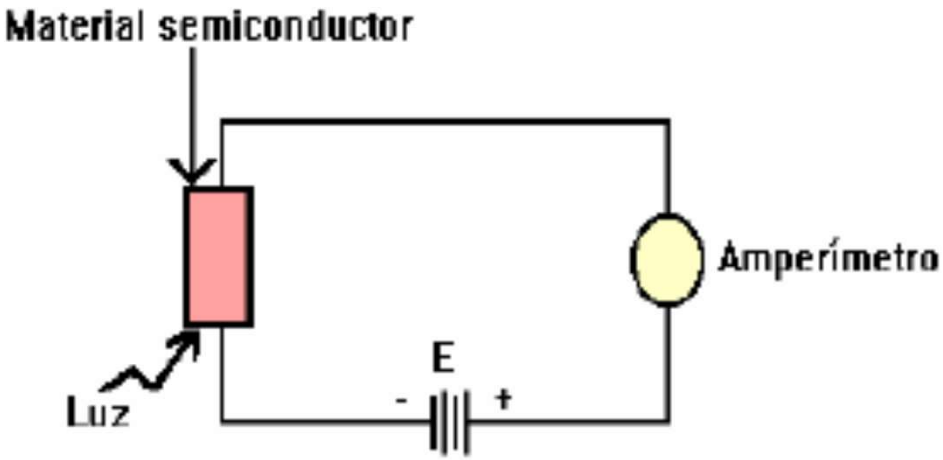


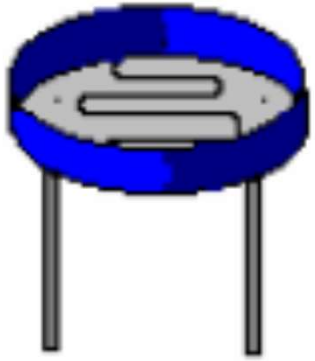


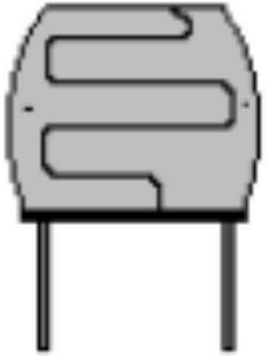
Sensores de Luz: Transductor Fotoeléctrico



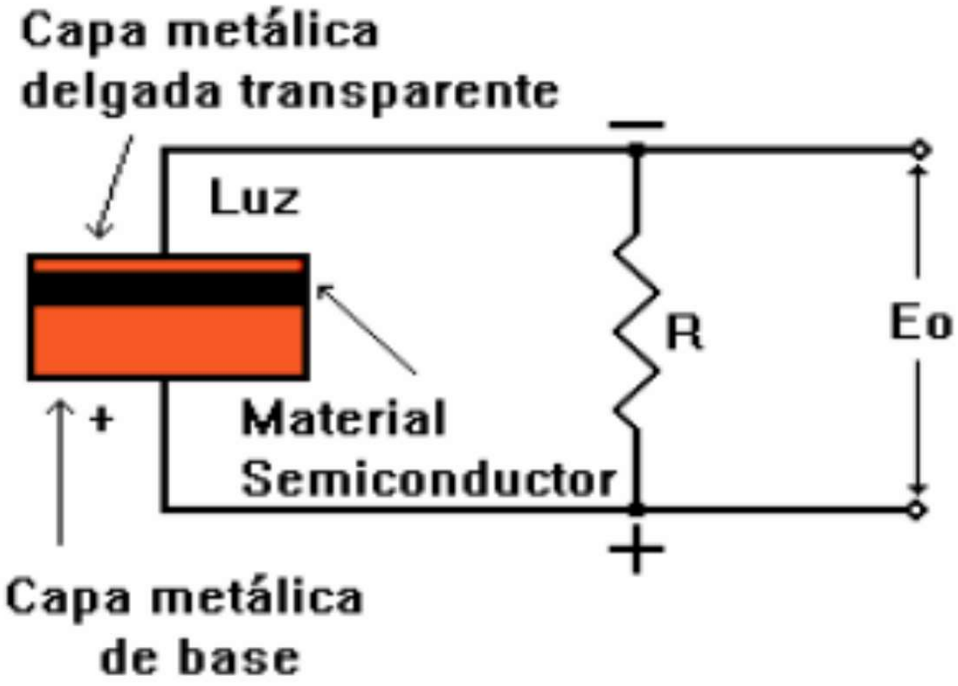
Sensores de Luz: Transductor Fotoconductivo



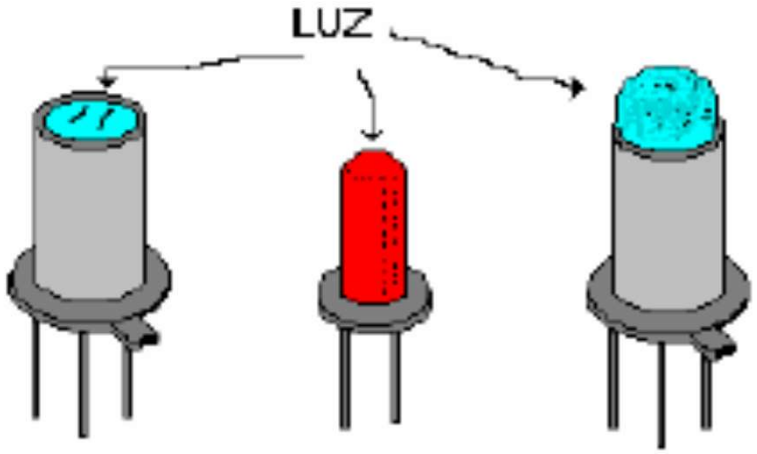


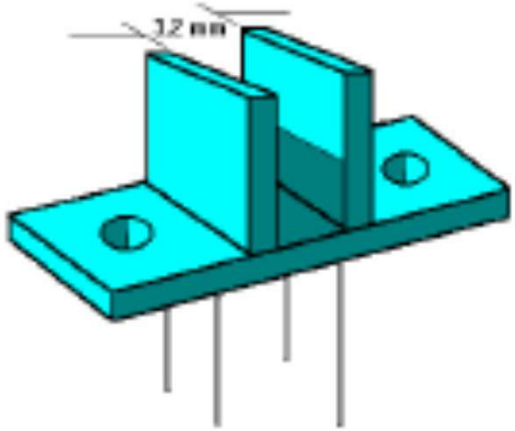


Sensores de Luz: Celdas Fotovoltaicas

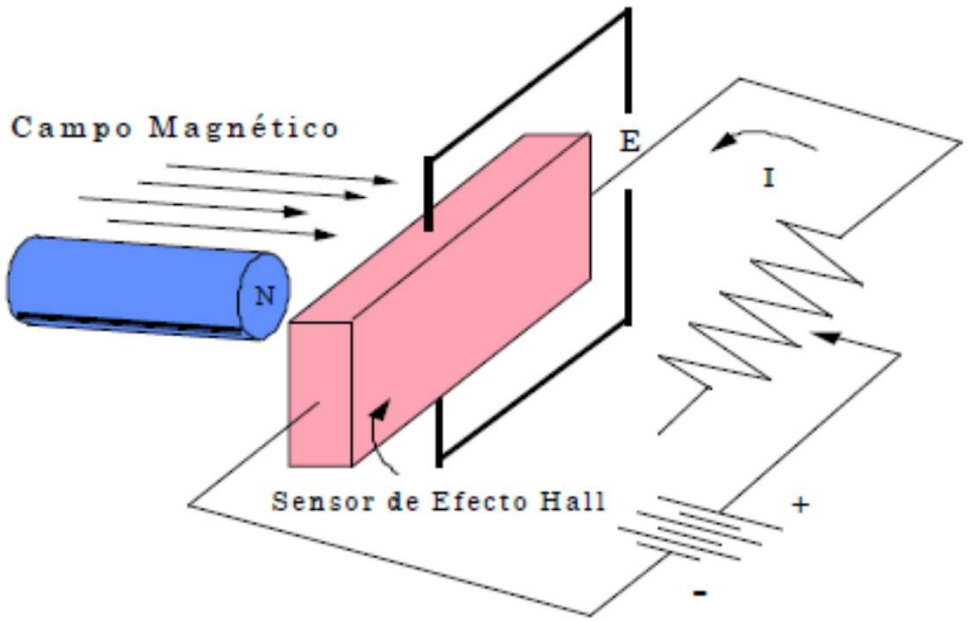


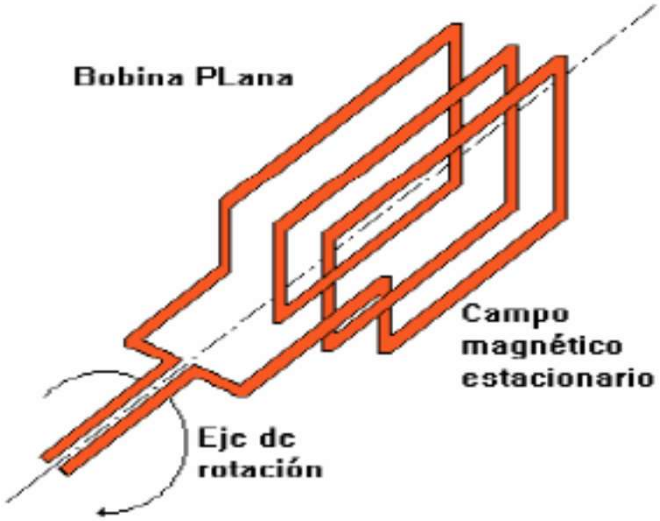
Sensores de Luz: Fotodiodos y Fototransistores



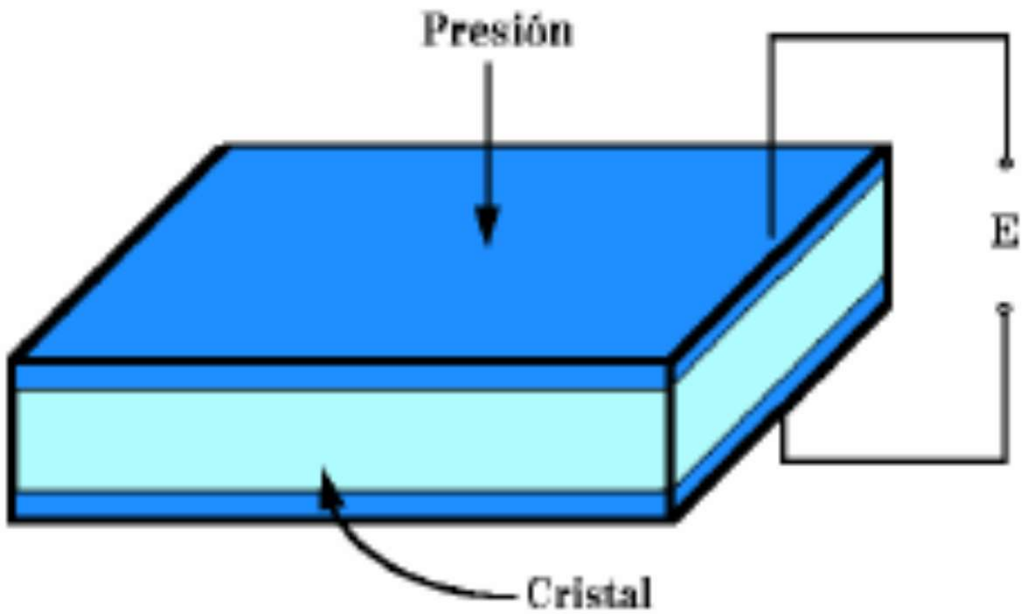


Sensores de Corrientes Eléctricas Continuas Altas

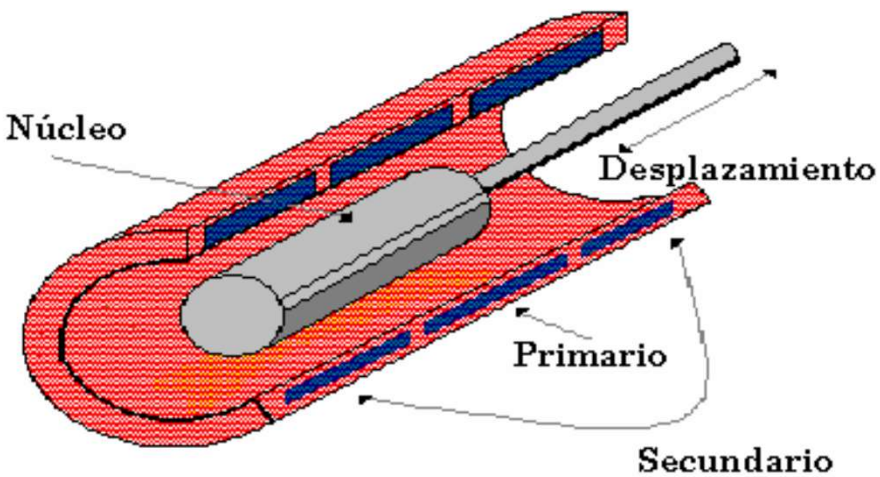


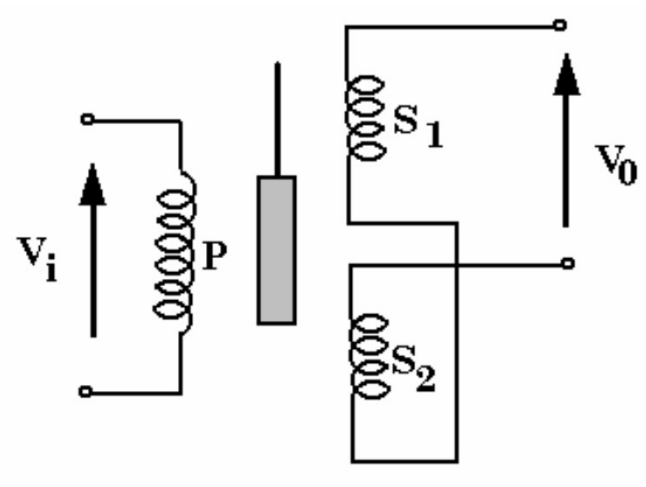


Sensores de Presión: Transductor Piezoeléctrico

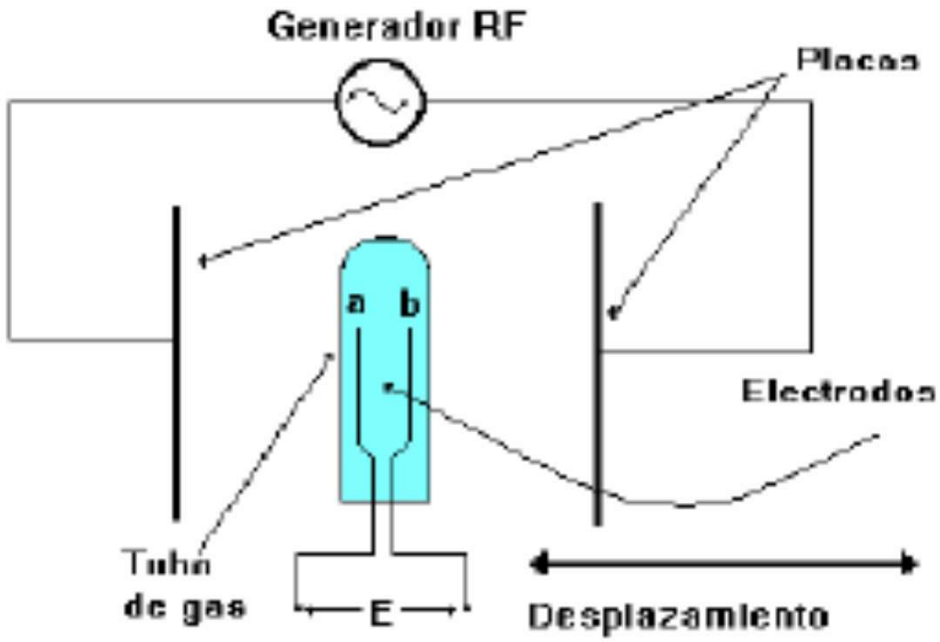


Sensores de Desplazamiento: Transductor con Transformador Diferencial

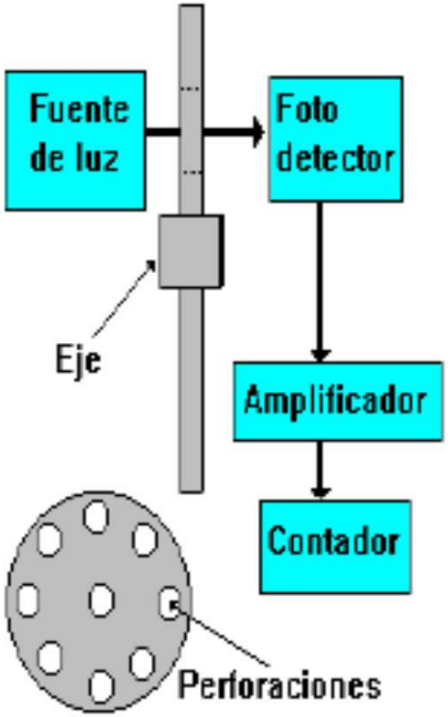


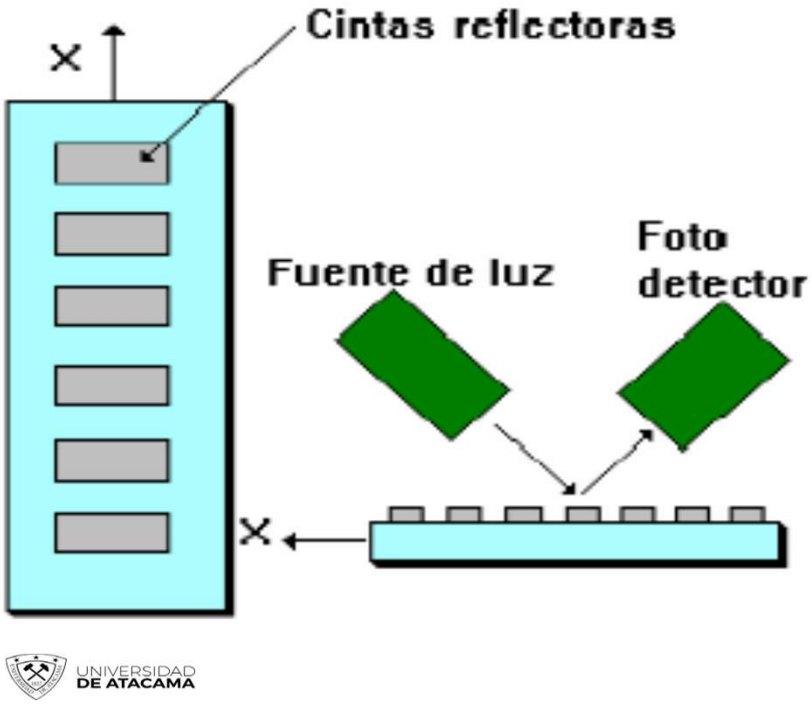


Sensores de Desplazamiento: Transductores de Ionización



Sensores de Desplazamiento: Transductores Digitales





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Profesor:
Nahur M. Meléndez Araya
Magister en Ingeniería Informática
nahur.melendez@uda.cl

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